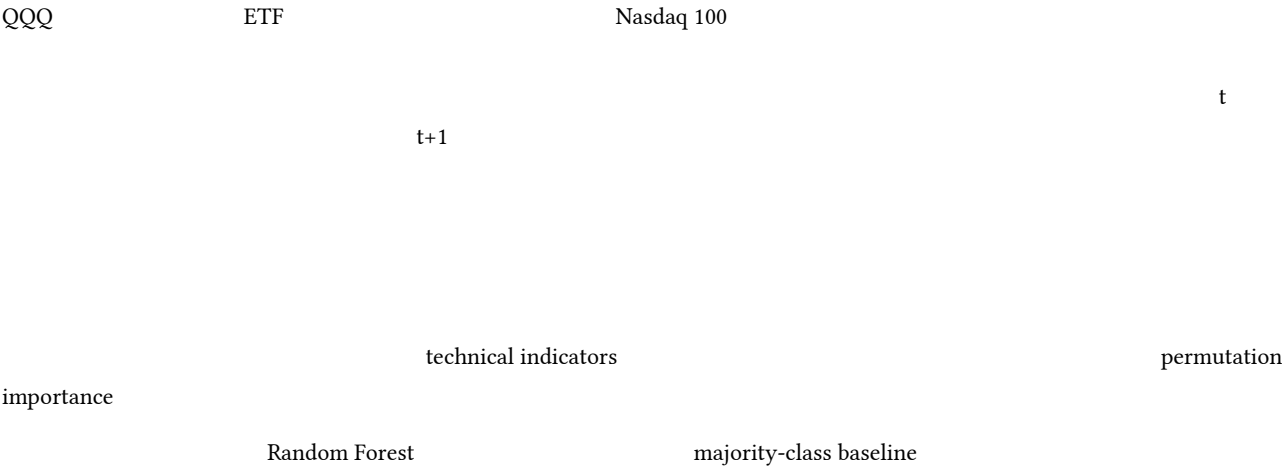


QQQ Technical Indicator Feature Importance

Big Data Analytics Project Report

Tien Hanprab		D6 900 557			
Nasdaq Historical Quotes for QQQ		4	2021	17	2026
26	2026				
					QQQ
technical indicators 21					
Nasdaq	1,433	1,412	[1]		
		283	Majority baseline	accuracy 54.77%	Random Forest
51.59%	HistGradientBoosting	53.00%	baseline		
accuracy					
Permutation importance	HistGradientBoosting	atr_14_pct	(	balanced accuracy 0.0315 ±	
0.0085	)				
notebooks/pipeline_qqq_5_year.ipynb					/ pipeline 5

1



RQ1 What are the major price, return, volume, and volatility patterns of QQQ during the configured analysis period?

RQ2 Which technical indicators are the most important features for classifying QQQ's next-day price direction?

RQ3 Does the Random Forest classifier perform better than a simple baseline on unseen chronological data?



2

QQQ Nasdaq public historical quote service endpoint
https://api.nasdaq.com/api/quote/XXX/historical assetclass=etf, fromdate=2021-01-01, todate=2026-09-17 limit=5000
Nasdaq XXX Historical Data [1] raw JSON metadata
response

	parse	
Date	datetime	
Open	float	USD
High	float	USD
Low	float	USD
Close	float	USD
Volume	integer	

Response 1,433 2021-01-04 2026-09-17 500
assignment [4] metadata retrieval 2026-09-19T22:15:36Z, URL SHA-256
5af8f404 421 142e426ca6 857 137b9355c4bee85fd8067ae6 923 973e3dab7001a
:
close response Nasdaq adjusted
close total return

3

Pipeline	schema	JSON	date	MM/DD/YYYY	datetime
/			OHLCV		
			missing	non-finite	
		High/Low			
Raw response	1,433	Nasdaq			
	0	missing OHLCV			
	0				
Clean OHLCV	1,433				
Indicator warm-up	20	20			
target	1	Close(t+1)			
Model-ready	1,412	21 features	next_return	target_up	

OHLCV

: 1,433 – 20 – 1 = 1,412

feature

warm-up

pipeline

audit

2021-02-02

2026-09-16

data/processed/5_year/qqq_features.csv

4

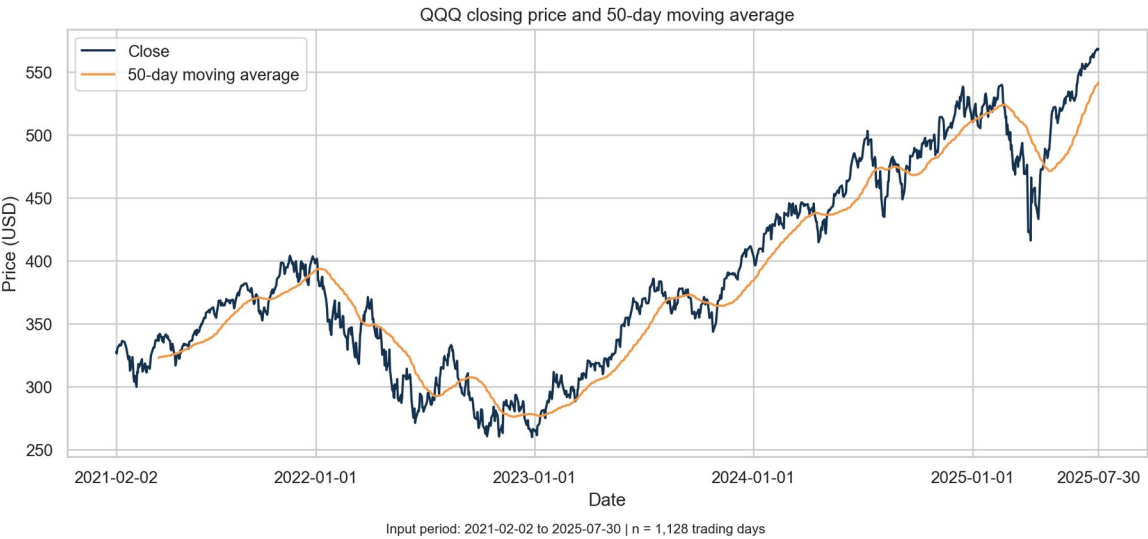
EDA

5-7

training period 2021-02-022025-07-30

1,128

holdout



1

50

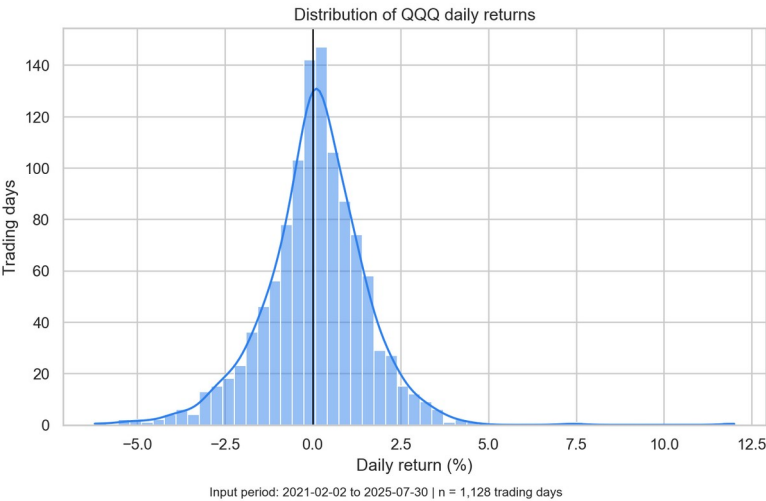
260.10

327.682022-12-28

568.02 USD568.14

(+73.35%)

2025-07-28

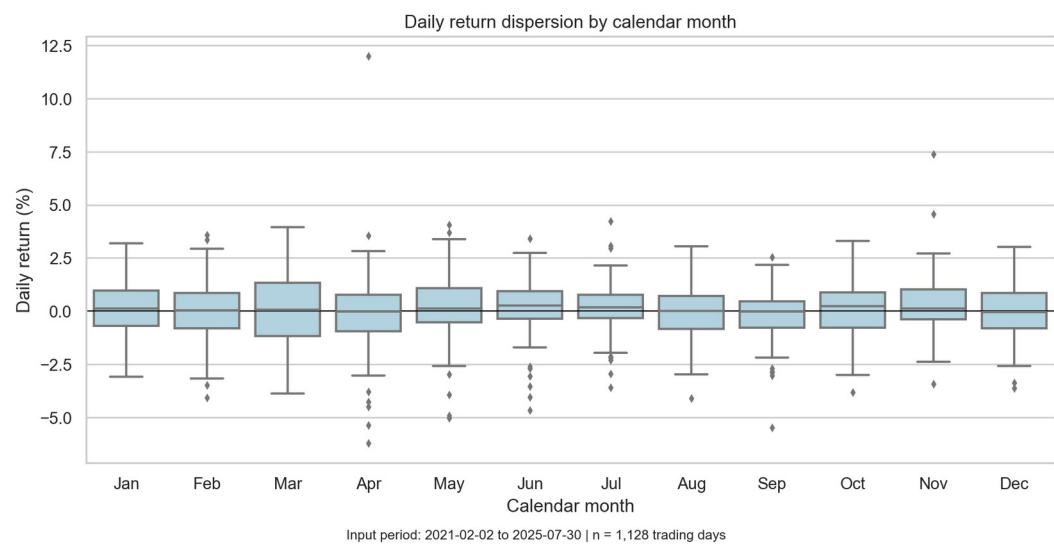


2

+12.0031%

0.0609%1.4621%—6.2109%

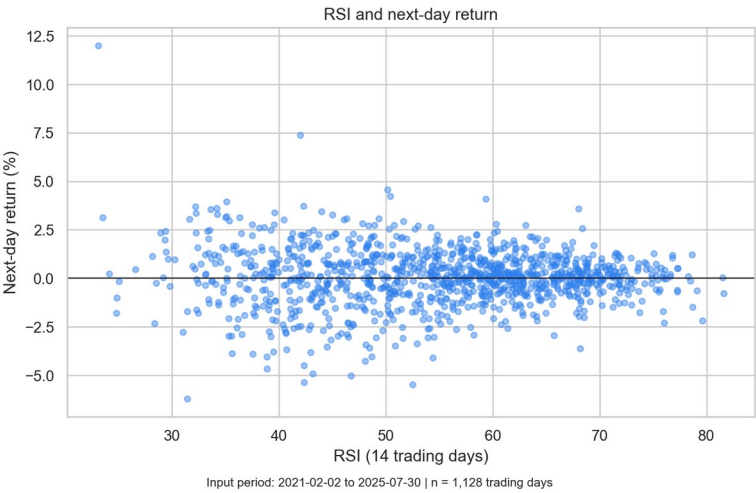
4



3 Box plot

Box plot

outliers



4 Scatter plot

RSI 14

RSI

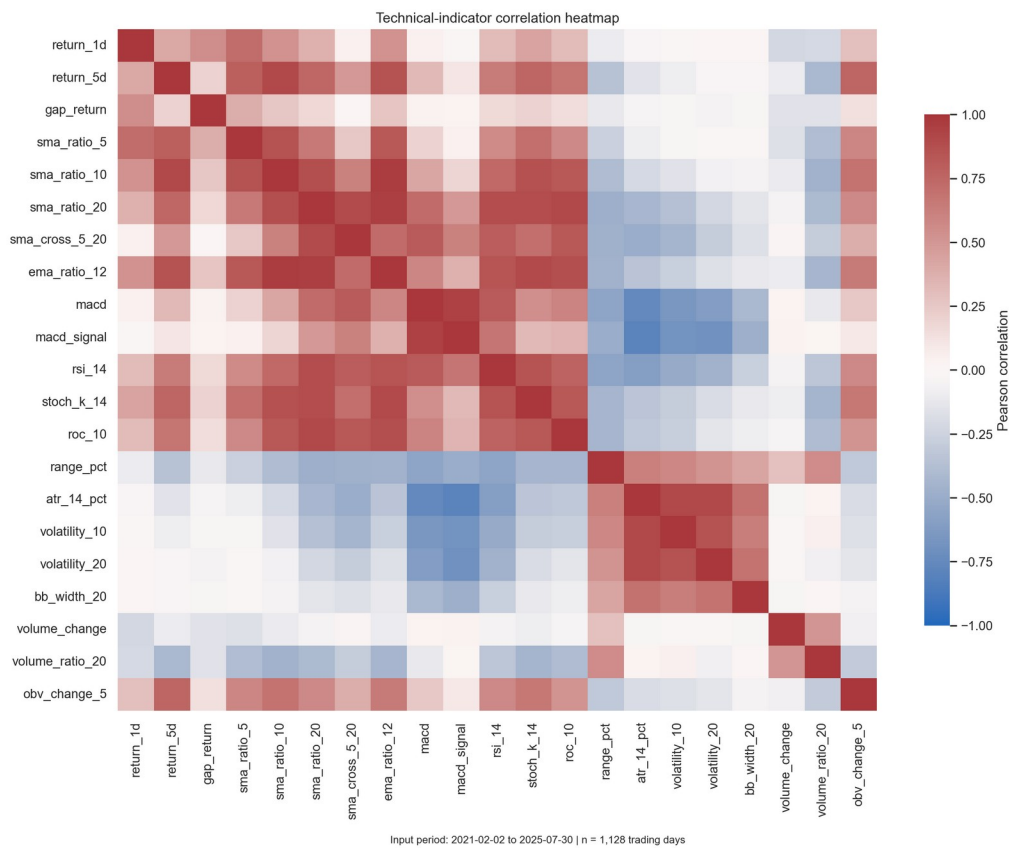
Pearson

RSI

next_return

-0.0312

4



5 Heatmap Pearson technical indicators

$r = 0.9506$ $r = 0.9694$ permutation importance [3]



6 Bar chart target training period

Up 614 (54.43%) Down or flat 514 (45.57%) balanced accuracy accuracy

5

Feature engineering 21 OHLCV t
return history 3, trend 7, momentum 3, volatility 5 volume 3 return_1d, SMA/EMA
ratios, RSI 14, ATR 14 volume ratio 20

Regression median imputation scaling Majority baseline , Logistic
min_samples_leaf 12, pipeline, Random Forest 700 max_depth 5
HistGradientBoosting 200 iterations learning_rate 0.03 output GradientBoosting

Train	1,128	2021-02-02	2025-07-30
Boundary gap	1	2025-07-31	
Test	283	2025-08-01	2026-09-16

train/test label
test train expanding TimeSeriesSplit 5 folds gap 1 [2] random
shuffle
accuracy, balanced accuracy, precision, recall, F1 ROC AUC test Permutation importance
GradientBoosting feature 30 test balanced accuracy
[3]

6

	Accuracy	Balanced	F1	ROC AUC
Majority baseline	54.77%	50.00%	70.78%	0.5000
Logistic regression	50.88%	50.06%	56.70%	0.4806
Random forest	51.59%	50.98%	56.51%	0.5034
GradientBoosting	53.00%	52.74%	56.39%	0.5269

RQ3: Random Forest

accuracy 51.59%

baseline 54.77%

3.18

balanced accuracy

Random Forest

50.98%

ROC AUC 0.5034

baseline

GradientBoosting

balanced accuracy

52.74%

accuracy 53.00%

baseline

1.77

ROC AUC 0.5269

Cross-validation

GradientBoosting

train

balanced accuracy

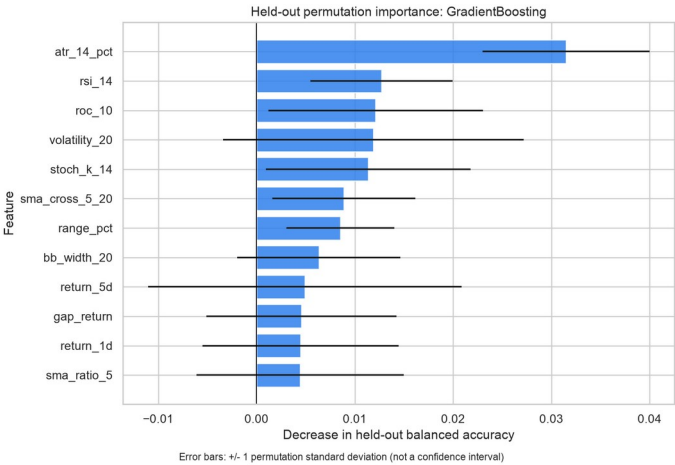
50.12% ± 2.37%

folds

ROC AUC

0.4965 ± 0.0115

holdout



7 Permutation importance

test

GradientBoosting

1 SD

RQ2: atr_14_pct

importance mean 0.0315 ± 0.0085

rsi_14 0.0127 ± 0.0072

roc_10 0.0121 ± 0.0109

volatility

importance 0.0577

importance

7

RQ1:	training					outliers
			49.89		EDA	
	RSI					
RQ2:	GradientBoosting	ATR 14		volatility	predictive association	
	permutation importance	test			metric	
RQ3: Random Forest		majority baseline	accuracy	283	GradientBoosting	balanced
accuracy	50%		CV	50%		
				walk-forward evaluation		
holdout		feature importance				
		adjusted/				
	OHLCV			Close		
	test				exploratory	

8

[1] Nasdaq. Invesco QQQ Trust Series 1 Historical Data. <https://www.nasdaq.com/market-activity/etf/qqq/historical> ; API request: <https://api.nasdaq.com/api/quote/QQQ/historical> (26 2026)

[2] scikit-learn developers. TimeSeriesSplit documentation. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.TimeSeriesSplit.html (26 2026)

[3] scikit-learn developers. Permutation feature importance documentation. https://scikit-learn.org/stable/modules/permutation_importance.html (26 2026)

[4] Big Data Analytics Project Assignment. [docs/Big_Data_Analytics_Project_Assignment.pdf](#)

outputs/figures/5_year/ : notebooks/pipeline_qqq_5_year.ipynb outputs/tables/5_year/; EDA